Sustainable Investing

Batteries: Key piece of the energy transition puzzle

SIGNATURE

Batteries are becoming critical to electrification, energy resilience and AI, as demand expands beyond EVs into storage, data centres and physical AI. We assess current and emerging battery technologies and identify public and private companies positioned across these evolving applications.

- **Battery demand is broadening beyond EVs as batteries become critical infrastructure for electrification, AI and strategic autonomy.** Passenger EVs remain the largest demand pool, but growth across power storage, data centres and physical AI is creating a more diversified demand profile and expanding batteries' role across the economy.
- **Performance requirements are fragmenting the market.** As demand expands beyond EVs, requirements are diverging across cost, energy density, safety, charging and duration, increasing the importance of chemistry selection and system-level engineering.
- **Next-generation technologies will complement rather than replace lithium-ion.** Solid-state, sodium-ion, flow, metal-air and lithium-sulfur batteries address different performance gaps, with adoption increasingly determined by application-specific requirements rather than a single winning chemistry.
- **Longer storage durations open the market beyond lithium-ion.** Li-ion remains highly competitive for shorter-duration storage, but its economics weaken as duration increases, creating opportunities for flow, metal-air, mechanical, thermal and other LDES solutions.
- **Regional policy is about competitiveness as well as resilience.** China sets the global cost curve through scale and integration, while US and European policy is driving reshoring but must overcome cost, execution and supply-chain constraints to generate returns.

Sustainable Investing Research

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- **As battery demand broadens across end-markets, value capture is unlikely to be uniform across the value chain.** We identify c.80 public and private companies that have exposure to areas where batteries solve a binding infrastructure constraint.

Executive summary

Batteries have moved from a cyclical EV component to a strategic infrastructure asset. The thesis around batteries has historically been heavily anchored to the electrification of road transport. This narrative is in the process of being rewritten, as developments in battery technology deliver rapid cost declines, expanding the breadth of economical use cases. Battery adoption now links utilities, power markets, autos, capital goods, data centre infrastructure, defence, mining, chemicals and industrial policy. This breadth makes batteries one of the few energy-transition themes that sits firmly at the intersection of decarbonisation, energy security and AI.

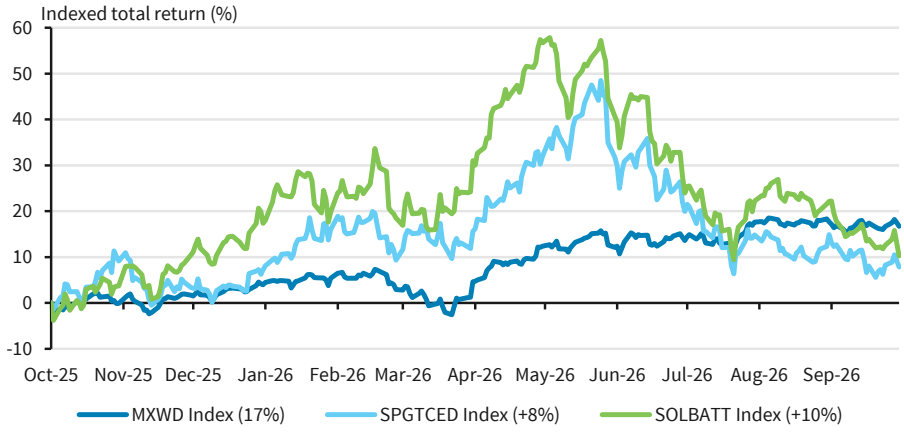
Batteries hold the keys to unlocking electrification, AI and strategic autonomy objectives. We see three megatrends converging to elevate the importance of batteries as critical infrastructure assets, providing a secular demand tailwind for companies in the value chain. Firstly, the green transition has increased the onus on batteries to provide grid flexibility and balance higher renewables penetration (Network security (part 1): Blackouts). Second, the AI revolution has accentuated the need for batteries to provide backup power power and peak shaving for power-hungry data centres (Data centres, renewables, the grid: positive or negative?). Finally, an increasingly volatile global geopolitical arena means batteries have become critical to physical AI deployment by unlocking runtime, mobility and resilience in defence tech (AI gets physical:The decade of the robot).

The demand stack has broadened faster than the market narrative. Global battery demand has risen from less than 200GWh in 2020 to almost 1,500GWh in 2025, with c.5,500GWh forecast by 2035.[1] While passenger EVs remains the largest demand pool today, growth over the next decade is expected to be driven as much by alternative drivers such as stationary storage, data centres and robotics. As such, we see a disconnect between this emerging reality and the prevailing market narrative, which still often frames batteries through a relatively narrow EV cost-curve lens.

Investing through the cycle. The batteries index (SOLBATT Index, +10%) has underperformed the broader market over the past year (MXWD Index, +17%), but has outperformed the clean energy index (SPGTCED Index (+8%) (Figure 1). It is worth noting that this is a relatively recent development, with the sector peaking at the start of June (+60%). The reversal since then has been driven by policy uncertainty and continued sensitivity to financing costs that have weighed on battery and renewable-energy company valuations, as the market priced in longer-term disruption in the Middle East and structurally higher energy prices, while the broader market benefited more from technology and AI-related growth.

[1] BloombergNEF. 2026 Long-Duration Energy Storage Outlook

FIGURE 1. Companies in the batteries value chain have underperformed the broader market over the past year



Source: Bloomberg, Barclays Research

Strategic autonomy is supportive, but not automatically profitable. China identified batteries as a strategic market more than a decade ago and now accounts for c.85% of global battery manufacturing capacity.[2] We estimate its share may fall toward c.60% by 2030 as reshoring efforts in the West take effect, but the cost gap remains material: European production costs are estimated c.50% above China, due partly to higher energy prices, higher environmental/safety requirements and skilled-labour shortages.[3] Western capacity announcements therefore need to be judged against commissioning, yield and cost, not nameplate capacity alone.

As battery demand broadens across end-markets, value capture is unlikely to be uniform across the value chain. As such, the ability to convert the demand stack into earnings through scale, supply chain flexibility, vertical integration and policy support capture will be key. We highlight companies that solve a binding infrastructure constraint below (Figure 2), which we profile in more detail in the attached excel document.

Batteries - Key piece of the Energy Transition puzzle - Company list.xlsx

Source: Company reports, Barclays Research

[2] IEA. Global Critical Minerals Outlook 2025

[3] International Energy Agency. The battery industry has entered a new phase

FIGURE 2. Battery technology developers across commercial and emerging solutions

Category	Technology	Public companies	Private companies
Current offerings: Battery technologies already in commercial production and supplied to customers	**Lithium-ion**	Amita Technology, BYD, CALB, CATL, CBAK Energy Technology, Dragonfly Energy, Enovix, EVE Energy, Farasis Energy, Ganfeng Lithium, Gotion High-tech, GS Yuasa, Leclanché, LG Energy Solution, Microvast, Murata Manufacturing, Panasonic Energy, Samsung SDI, SK On, Sunwoda, Tesla, REPT Battero, Saft, Kokam, FDK, Maxell, Veken Technology, VITZROCELL, Toshiba Battery, Ford, General Motors, BMW and Eni	ACC (Automotive Cells Co.), AESC, Clarios, Our Next Energy (ONE), Verkor, SVOLT Energy, HiTHIUM and CEGASA
	Solid-state and semi-solid-state	Ganfeng Lithium, Maxell	-
	Sodium-ion	CATL, Farasis Energy, ESS Tech	HiTHIUM and Clarios
	Long-duration energy storage (LDES)	ESS Tech, Eos Energy Enterprises, Invinity Energy Systems and Sumitomo Electric Industries	Form Energy, EnerVenue, VFlowTech and CEGASA
In development: Emerging battery technologies at development, pilot or early-commercial stage	**Solid-state and semi-solid-state**	Blue Solutions (Bolloré Group), FinDreams Battery (BYD), CALB, CATL, Factorial Energy, Farasis Energy, LiEnergy / FunLithium (Ganfeng Lithium), Gotion High-Tech, GS Yuasa, Leclanché, LG Energy Solution, Maxell, Panasonic Energy, QuantumScape, Samsung SDI, SK On, Solid Power, TDK, Ilika and BMW	24M Technologies, AESC, Basquevolt, ION Storage Systems, ProLogium Technology, SVOLT Energy and WeLion New Energy
	Lithium-metal, lithium-sulfur and lithium-air	Blue Solutions (Bolloré Group), CALB, LiEnergy / FunLithium (Ganfeng Lithium), GS Yuasa, LG Energy Solution, QuantumScape, Samsung SDI, SES AI and Solidion Technology	Basquevolt, Lyten, PolyPlus Battery Co., Sion Power, Theion and Zeta Energy
	Next-generation lithium-ion cells and anodes	Amprius Technologies, CATL, EVE Energy, Leclanché, Murata Manufacturing, Panasonic Energy and Solidion Technology	ACC (Automotive Cells Company), 24M Technologies, AESC, Anthro Energy, Group14 Technologies, Nanotech Energy, Nyobolt, Our Next Energy (ONE), Sila, StoreDot and Verkor
	Sodium-ion	ESS Tech, FinDreams Battery (BYD), CATL, EVE Energy, Farasis Energy, LG Energy Solution, Varta and General Motors	Altris, Faradion (Reliance Industries), Peak Energy, Tiamat Energy and AESC
	LDES and other emerging chemistries	ESS Tech and Eos Energy Enterprises	Form Energy, Group1, Energy Dome, and Hydrostar

Source: Company reports, Barclays Research

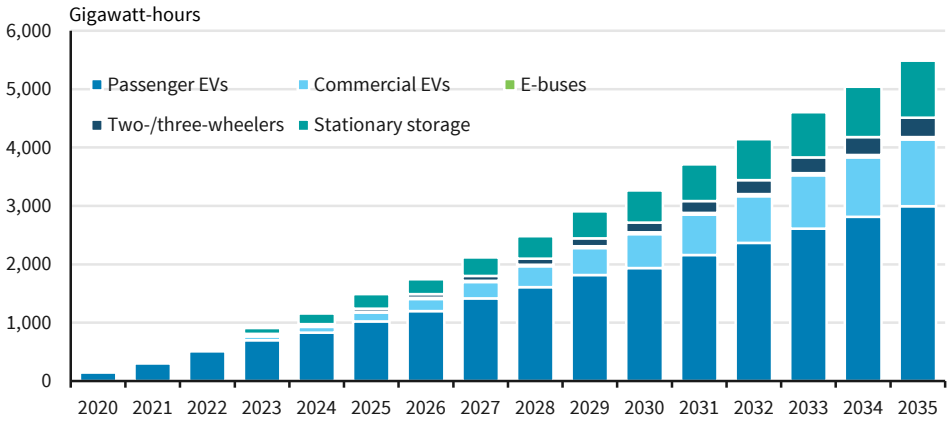
Batteries: Critical enablers of electrification, AI and autonomy

Demand diversification set to drive threefold increase in deployment by 2035

The battery market has displayed signs of exponential demand growth since 2020, as declining cost curves and rising electrification momentum take shape. Global battery demand amounted to less than 200GWh as recently as 2020, rising to just under 1,500GWh in 2025, implying a CAGR of c.50%. The main component of the battery demand stack has historically been EVs, currently accounting for c.70% of demand, although demand drivers have begun to diversify in recent years (Figure 3).

This broadening of the battery demand drivers underscores the sector's positive growth prospects. Alternative demand pools such as commercial EVs, E-buses and stationary storage are expected to grow from 30% of demand today to nearer 50% by 2035.[4] Taken together, these sectors are expected contribute an additional 2,000GWh by 2035, similar to that of passenger EVs.[5] As such, our base case is that battery demand growth maintains an elevated albeit lower growth rate over the next decade, converging on a CAGR of c.15% towards BloombergNEF's 2035 demand forecast of c.5,500GWh.[6]

FIGURE 3. Global battery demand is projected to triple by 2035, driven by a broadening demand pool to sectors such as stationary storage and commercial EVs



2026-35 are BloombergNEF forecasts.
Source: BloombergNEF, Barclays Research

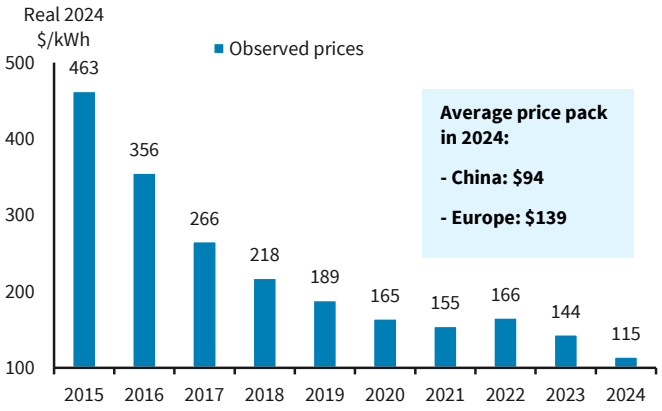
Growing demand unlocked by a step-change in lithium-ion technology

Lithium-ion batteries have emerged as the dominant battery chemistry, accounting for ~90% of worldwide battery capacity deployment. They have a range advantage over legacy technologies: higher energy density, lower weight, the ability to be recharged multiple times with limited degradation, as well as minimal energy loss when not in use. Coupled with rapidly declining production costs, this has seen lithium-ion batteries reshape industrial and energy-system applications as well as becoming ubiquitous in consumer electronics and transport technology.

[4] BloombergNEF. 2026 Long-Duration Energy Storage Outlook
[5] BloombergNEF. 2026 Long-Duration Energy Storage Outlook
[6] BloombergNEF. 2026 Long-Duration Energy Storage Outlook

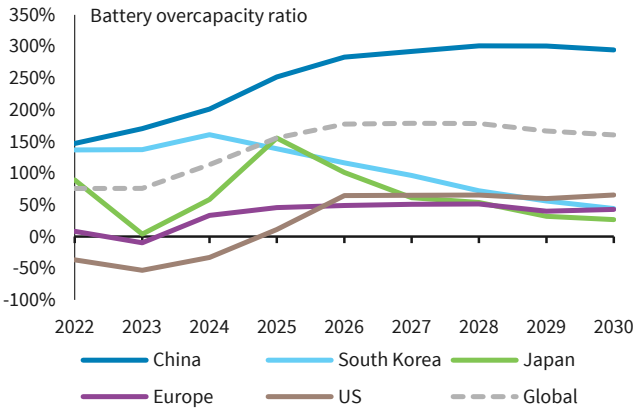
Lithium-ion technology is therefore at the centre of the battery growth story, with deployment having grown sixfold since 2020. This has been enabled by technological improvements, which have brought down production costs by 75% in under a decade, shifting the economics of electrification and incentivising greater uptake (Figure 4). Chinese overcapacity has been the main driver for these cost declines, creating significant pricing pressure for Western producers (Figure 5).

FIGURE 4. Lithium-ion batteries prices have contracted sharply over the past decade...



Source: Bloomberg, S&P Global Mobility (Nov.25), ICCT, Barclays Research

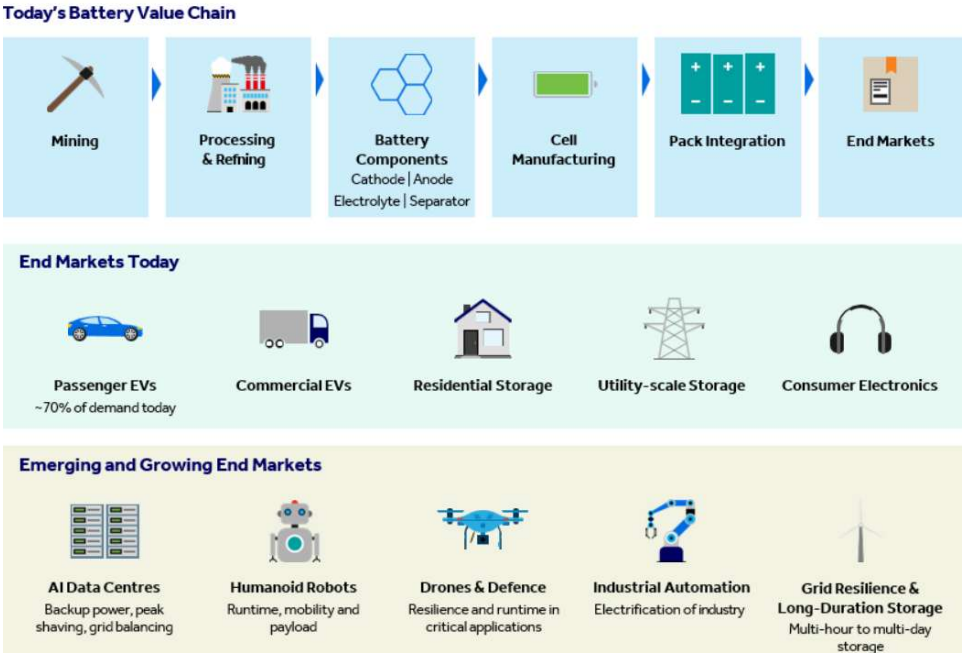
FIGURE 5. ...driven largely by Chinese overcapacity



Source: China customs, Wind, Bloomberg, Barclays Research

Alongside lithium-ion, legacy battery technologies remain in extensive use. Lead-acid batteries continue to play a critical role in automotive starting systems, backup power and industrial applications due to their low cost, reliability and mature recycling infrastructure. For smaller consumer devices, alkaline batteries are still widely used where low cost and long shelf life matter more than rechargeability, while nickel-metal hydrid batteries still have a niche presence in certain rechargeable household and industrial uses.

FIGURE 6. The battery value chain and primary uses of batteries



Source: Barclays Research

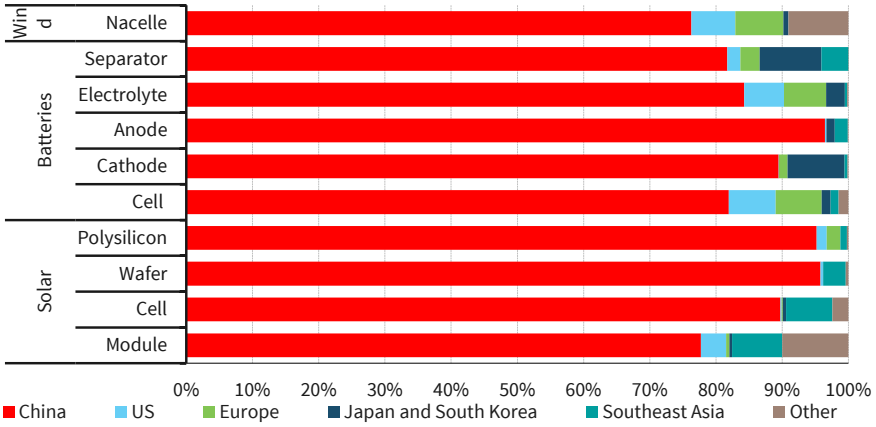
China dominates, but Europe and US reshoring efforts closing the gap

Chinese comparative advantages is well established...

China currently dominates the global battery industry, accounting for c.85% of cell manufacturing worldwide. As we discussed in China's green transition: gains and opportunities, China has cemented a manufacturing comparative advantage across the broader clean energy technology stack (Figure 7), as well as upstream minerals mining and refining (Transition minerals: unearthing opportunities from a $500bn supercycle). This is the outcome of a multi-decade industrial strategy, which has prioritised growing Chinese producers' market share above profitability through coordinated policy action across both national and local government, ranging from concessional loans, domestic contents-conditioned tax breaks and subsidies.

We see China's lead in batteries as entrenched in the near to medium term (pre-2035). The cost advantage that Chinese battery plants have accrued owing to a supportive policy environment coupled with favourable economies of scale and lower labour costs is unlikely to be reversable over the next decade, in our view. The upshot is that Chinese producers are likely to remain the benchmark against which competitors in the West will be required to compete on cost, speed and supply-chain depth. Nevertheless, we expect China's share of battery manufacturing to decline over the next decade to levels nearer 60% driven as much by the CCP's anti-involution policy geared towards curbing 'excessive low-price competition' (China: Anti-involution revisited), with the CCP having announced plans to phase-out battery export AVT rebates by 2027[7], as by Western policymakers' efforts to reshore manufacturing value chains (Strategic autonomy's the name of the game).

FIGURE 7. China dominates upstream mining and refining of mineral inputs into batteries, cementing the advantage of its domestic battery manufacturers



Source: BloombergNEF

...but Europe and US reshoring efforts are gathering steam

In our view, Western battery manufacturers are unlikely to compete on costs in the near term. However, protectionist policy measures deployed by European and US governments will reduce Chinese competition and alleviate margin compression pressures. Policymakers in both Europe and the US have identified an over-reliance on Chinese exports as a security and competitiveness issue, although they diverge materially in policy prescription. While Europe has implemented more of a 'stick-based' regulatory approach through the imposition of

[7] Bloomberg. China to Cut Export Tax Rebates to Ease Global Trade Tensions

sustainability requirements, the Trump Administration has adopted protectionist measures designed to shield US industry through domestic content requirements. **These approaches should enable western battery companies to grow meaningfully by servicing a greater share of domestic demand.**

The EU is attempting to build out its domestic battery manufacturing capacity, having doubled down on a green industrial strategy approach since the Covid-19 pandemic in order to meet domestic competitiveness, decarbonisation and energy security objectives. The bloc has imposed strict sustainability requirements for battery manufacturing, which Chinese firms will likely struggle to meet, generating a carbon comparative advantage for European manufacturers. The EU's Batteries Regulation is the centre piece of the bloc's battery strategy, setting lifecycle-wide requirements covering sustainability, safety, carbon footprint disclosures, circularity and due-diligence on raw-material sourcing, in addition to mandatory digital battery passports for EV and industrial batteries. Meanwhile, rapid renewables and EV penetration in Europe has created strong demand for batteries, with the EU Industrial Accelerator Act setting out incentives for domestic battery manufacturing capacity to meet this demand (Sustainable Investing: Clean and competitive). The EU is also attempting to reshore/ friendshore upstream mining and processing through the Critical Raw Materials Act, which should support its domestic clean tech manufacturing base. Despite these initiatives, there remain material bottlenecks to mass deployment, with 600GWh of European battery manufacturing capacity currently stalled or delayed (c.35% of the bloc's total capacity)[8] and production costs remaining above those of Chinese competitors by c.50%.[9]

The Trump Administration's strategic approach to the battery sector is bifurcated, straddled between its push to reindustralise the US, while domestic clean tech manufacturing and demand subsidies are simultaneously de-prioritised. On the supply side, it has deployed an aggressive industrial strategy dating back to the Biden Administration, which has sought to reshore and strengthen its domestic minerals sector, while largely maintaining IRA manufacturing tax credits for domestic battery producers, provided they meet tight foreign entity of concern (FEOC) rules.[10] On the demand side, however, it has removed large swaths of consumer tax credits and incentives for EVs and residential solar paired with batteries (Taking stock of US power post-OBBB). While IRA tax credit effectively subsidises each domestically produced kWh of capacity by $35, the stringent nature of FEOC rules, coupled with the high dependence on Chinese battery inputs, makes the impact more likely to be muted in practice. This seemingly contradictory approach suggests to us that the administration is likely to support the growth of the domestic battery sector, but will seek to steer it more towards alternative use cases such as AI data centres and humanoid robots, rather than clean tech deployment. As such, we expect to see domestic US battery manufacturing increase from current levels of c.5% of global production but remain below the ~15% market share that bodies such as the DOE had predicted the US market would accrue before the OBBBA amendments.[11]

[8] Interact Analysis. Problems hamper over 30% of European Li-Ion battery projects

[9] IEA (2025). The battery industry has entered a new phase

[10] FEOC restrictions require 60% of qualifying battery inputs to come from non-Chinese sources by 2026, rising to 85% by 2030

[11] Department of Energy. 2021-2024 Four-year Review of Supply Chains for the Energy Sector Industrial Base

FIGURE 8. US, China and EU: Battery market comparative analysis

Region	Current market position	Policy initiatives	Comparative advantage	Investor implications
China	Dominant across the value chain: ~85% of global battery manufacturing capacity today; expected to remain the largest single player even as share declines toward ~60% by 2030.	Long-standing industrial policy (e.g. 5-Year Plans, provincial support), coordinated supply chain build-out, and export-led scale advantages.	Cost leadership + full supply chain integration (refining, cathodes/anodes, cells, manufacturing). Strong position in LFP and commoditised chemistries.	Benchmark price setter globally; margin pressure for non-Chinese producers. Best positioned for volume growth but not necessarily value capture (commoditisation risk). Equity exposure favours scale leaders and low-cost operators; bear case for high-cost Western capacity.
United States	Smaller starting base (~low single-digit share historically), but scaling rapidly with new capacity announcements and reallocation toward storage and domestic supply.	IRA-led incentives (e.g. Section 45X production credits ~$35/kWh for cells) and storage credits (48E), combined with supply-chain localisation and China-exclusion rules under FEOC rules.	Policy-driven localisation + end-market demand pull (EVs, storage, data centres). Increasing role in energy storage manufacturing and deployment.	Returns depend on execution and policy capture. Attractive where companies combine subsidy eligibility + scale + technology differentiation. Risk that capacity comes online without cost competitiveness, thus project execution, utilisation and margins critical.
Europe	Moderate demand base, but structurally behind on cost and scale; production costs estimated ~50% higher than China, with delays in project buildout.	Regulatory-led framework (EU Batteries Regulation, Net-Zero Industry Act, Critical Raw Materials Act) focused on sustainability, traceability and localisation.	Regulation, premium demand and sustainability-led differentiation (e.g. battery passports, lifecycle standards). Strong industrial and automotive demand base.	More favourable for high-value segments (premium chemistries, ESG-compliant supply chains, specialised applications). However, weaker economics for commodity cells create risk of stranded or underutilised capacity. Selectivity therefore critical by avoiding volume-only plays.

Source: Barclays Research

Diversifying demand pool underscores bullish thesis

Our bullish view is anchored in three structural demand forces that are expanding the battery market beyond a single end-market cycle. These are:

- The EV revolution going global
- Long-duration storage: the resilience layer that enables electrification and AI-load growth
- Physical AI: widening the battery demand stack during the Decade of the Robot

FIGURE 9. Battery demand drivers and implications for investors

Structural demand driver	Commercial thesis	Battery relevance	Investment implications
The EV revolution: going global	EVs remain the core demand driver, with growth widening across regions, vehicle classes and commercial fleets.	Batteries remain the key constraint to EV scale. Commercial fleets increasingly favour chemistries and suppliers positioned for cycle life, safety and lower total cost of ownership.	Beneficiaries include scaled OEMs, cell makers and materials suppliers with cost leadership, yield gains and chemistry flexibility; in commercial transport, also watch fleet electrification, charging and LFP-linked supply chains.
Long-duration storage: the resilience layer that enables electrification and AI-load growth	Battery storage is moving from a supporting technology to essential grid infrastructure as renewables rise and reliability becomes more valuable.	Stationary storage is a major incremental battery demand pool, supported by grid deployment, backup power and data-centre-related load balancing.	The strongest monetisation may be downstream: utilities, BESS integrators, power equipment vendors, software providers and bankable project platforms.
Physical AI: widening the battery demand stack during the Decade of the Robot	AI is extending into robotics and automation, creating new battery demand where runtime, charging speed, safety and integration matter.	Batteries are the local energy backbone for robots and autonomous systems. The market is smaller today but structurally additive and more performance sensitive.	Advanced chemistries, thermal management, power electronics and automation platforms stand to benefit.

Source: Barclays Research

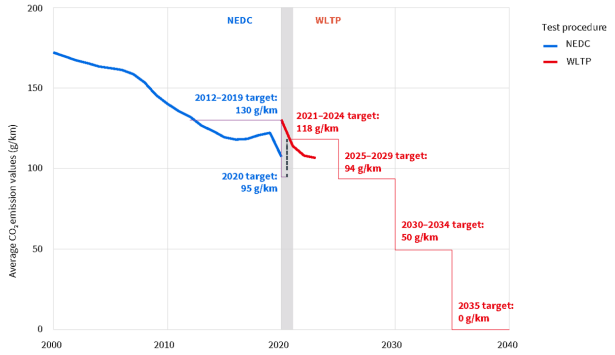
The EV revolution: Going global

Battery producers are set to see their main demand driver continue to grow over the coming decades, with EV penetration ramping up outside of the established markets of China and Europe. While the market narrative around EVs has shifted from one of hypergrowth to a more two-speed EV cycle by region, we see the compelling economics of road transport electrification as a secular driver of global EV adoption. Global EV penetration in passenger vehicles has reached c.25% globally, but this average masks important regional disparities, with half of new vehicle sales in China currently EVs. We see battery demand being lifted in this segment of the market, as global passenger EV penetration converges towards 50% in the 2030s. This is further supported by the maturation of commercial EVs, with battery demand from this segment of the market expected to rise roughly sevenfold by 2035[12], reflecting the ameliorating economics of high-utilisation fleets, logistics and public transport.

We see the European EV market as a strong growth market for domestic battery manufacturers. Environmental regulation provides a long-term support for the demand floor, particularly in Europe (Figure 10), forcing a ramp-up in EV penetration as manufacturers seek to meet emissions targets and mitigate the risk of financial penalties (Figure 11).

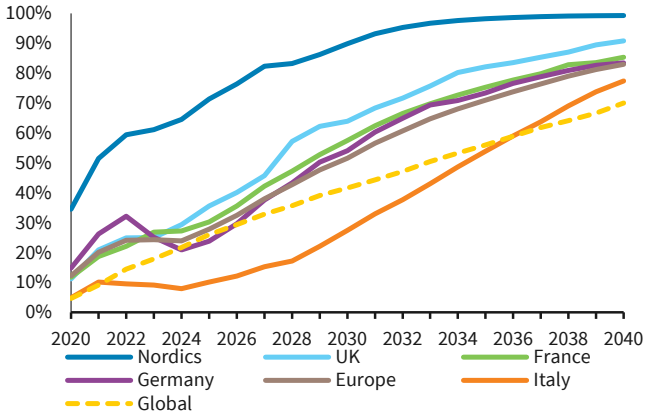
[12] BloombergNEF. 2026 Long-Duration Energy Storage Outlook

FIGURE 10. CO2 emission regulations becoming more stringent in Europe...



Source: ICCT, Barclays Research

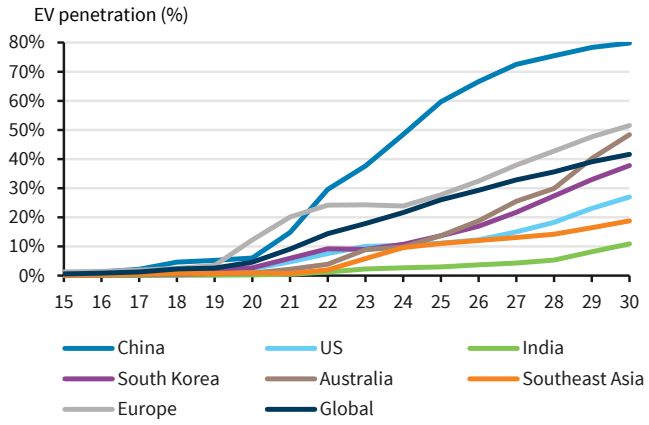
FIGURE 11. ...driving higher required BEV share of sales for auto makers to meet C02 targets



Source: ICCT, Barclays Research

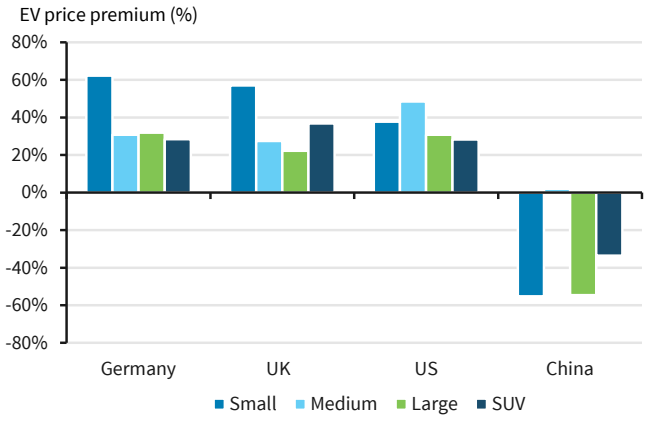
EV penetration growth is also accelerating in EMs, as Chinese automakers ramp up sales outside their domestic market. Australia, India and SouthEast Asia in particular are expanding the Total Addressable Market (TAM), reinforcing global battery demand (Figure 12). However, affordability is emerging as a near-term constraint, particularly in Europe and the US (Figure 13). Our Autos colleagues highlight that new vehicle orders and production are under pressure in 2026, reflecting tariffs, weak consumer confidence and affordability issues caused by higher vehicle prices, tighter financing conditions and high vehicle financing rates. We still see these headwinds as being offset by the expanding TAM of passenger EVs. As such, it is worth noting that the largest current end-market for batteries remains exposed to the consumer cycle, although the long-term structural demand story remains positive.

FIGURE 12. EV penetration is projected to ramp up beyond the current established markets of China and Europe...



2026-30 are BloombergNEF forecasts.
Source: BloombergNEF, Barclays Research

FIGURE 13. ...however, affordability is a growing issue in the West



Source: BloombergNEF, Barclays Research

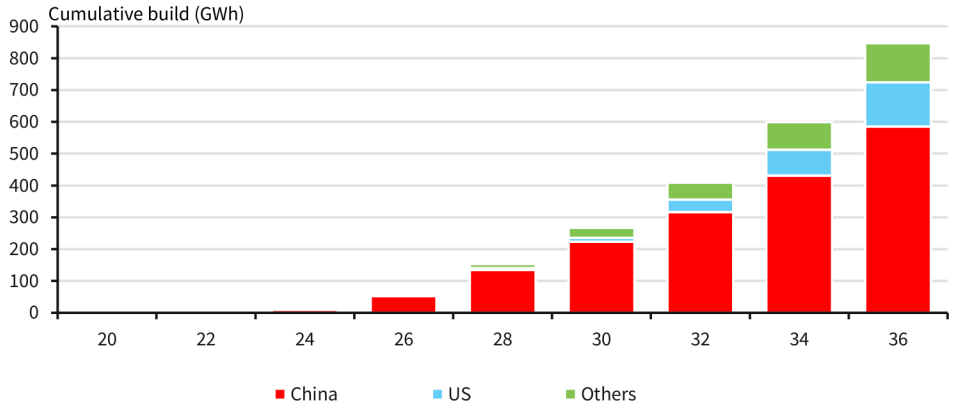
This makes chemistry mix central to the autos demand story. Lithium Iron Phosphate (LFP) batteries' lower cost, safety and cycle-life advantages are already relevant for commercial EVs and stationary storage, but they also matter for passenger EV affordability. In an environment where OEMs face pressure to scale BEVs while managing margins, frontier

chemistries are emerging as a route to make EV demand more durable and ensure competitiveness with established Chinese OEMs.

Long-duration storage: The resilience layer for electrification and AI-load growth

Long Duration Energy Storage (LDES) is emerging as a core infrastructure asset which serves to stabilise the grid stress associated with higher variable renewables penetration and AI data-centre load growth. This trend is supported by the shift towards longer-duration storage beyond standard two-hour configurations, making stationary storage a key resilience asset rather than just a short-term grid cycle stabiliser.

FIGURE 14. LDES capacity deployment is projected to ramp up over the next decade



2026-36 are BloombergNEF forecasts.
Source: BloombergNEF, Barclays Research

The rapid growth in AI data centre load growth has made Powering AI not only a generation but also a storage story. The primary use of batteries in data centres is the uninterruptible power supply system (UPS). These provide backup power during blackouts or grid disturbances and can help bridge the energy requirements before back up generators kick in. Regional deployments of batteries are in line with data-centre build-outs. North America remains a key market, driven by the high concentration of data centres and continued expansion plans by hyperscalers to meet rising AI related demand, which is expected to further increase battery deployment. Europe is also seeing more data centres constructed and an increase in battery deployment supported by concerns around grid stability (e.g. Spain blackout) and higher renewable energy integration. Analysis we published with our European Utilities colleagues shows that every GW of data-centre power demand could require 6-10GW of solar or 1-3GW of offshore wind capacity, underscoring the importance of LDES for backup power, grid balancing, peak shaving and energy management (Data centres, renewables, the grid: positive or negative?).

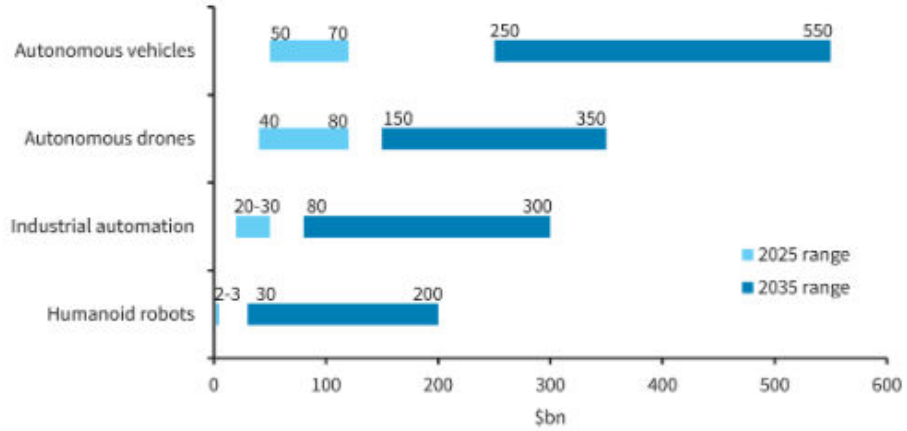
Physical AI: Widening the battery demand stack during the Decade of the Robot

Physical AI extends the battery story from power systems into the real economy. As discussed in AI gets physical: The decade of the robot, batteries are one of the core '3Bs', alongside Brains, Brawn and Batteries, underpinning robotics, autonomous vehicles and drones by providing the local energy backbone for these systems.

Deployment is already inflecting. We estimate that c.15k humanoid robots were installed globally in 2025 (up from c.2k in 2024) and forecast c.60k deployments in 2026, with China accounting for ~85% of installations. This scaling reflects a broader shift toward embodied AI

systems operating across manufacturing, logistics and services, creating a new layer of battery demand that is structurally different from EVs or grid storage. Although order volumes remain small today, our colleagues forecast significant growth across all the major verticals (Figure 15). Moreover, while volumes are smaller today, the specific technical requirements driving demand (energy density, safety, fast charging, BMS performance) are more stringent and less price-driven, which may open the door for new specialised players in Europe and the US (AI gets physical: The decade of the robot).

FIGURE 15. **Physical AI market size, 2025 and 2035 projections**



Ranges are Barclays Research estimates based on Alphasense and company data.
Source: Company reports, AlphaSense, Barclays Research

The increasing importance of physical AI in defence reinforces the diversification benefit of new sources of demand, with clearer willingness to pay for capability over cost. In physical AI systems such as drones, batteries directly determine flight time, payload capacity and operational range, making them a core determinant of mission effectiveness rather than a commoditised input. Batteries now power soldiers' equipment, vehicles and satellites, enabling military units to be more mobile and resilient without constant access to fuel or grid power. As such, defence demand is strategically important given it introduces policy-supported price-inelastic demand, which supports higher-spec chemistries and advanced battery management systems.

Beyond volume growth, we think physical AI could also drive a shift in how battery performance is valued. While much of the industry focus has historically centred on energy density and cost reduction, the economics of robot fleets increasingly depend on utilisation rates and application-specific performance requirements. For example, a humanoid robot that operates for two hours but requires an hour to recharge may only be productive for two-thirds of a working day. Faster charging could therefore become as important as extending runtime, allowing operators to increase productive hours without expanding fleet sizes. Similarly, battery requirements are becoming increasingly differentiated across physical AI applications. Humanoid robots require a balance between runtime, safety and charging speed, autonomous vehicles are likely to prioritise cycle life and durability, while drones place a premium on reducing weight to maximise payload capacity and range. As a result, future battery demand may not be fully served by standard EV or consumer-electronics batteries, creating opportunities for specialised chemistries, cell formats and battery management systems tailored to the specific operational requirements of physical AI systems.

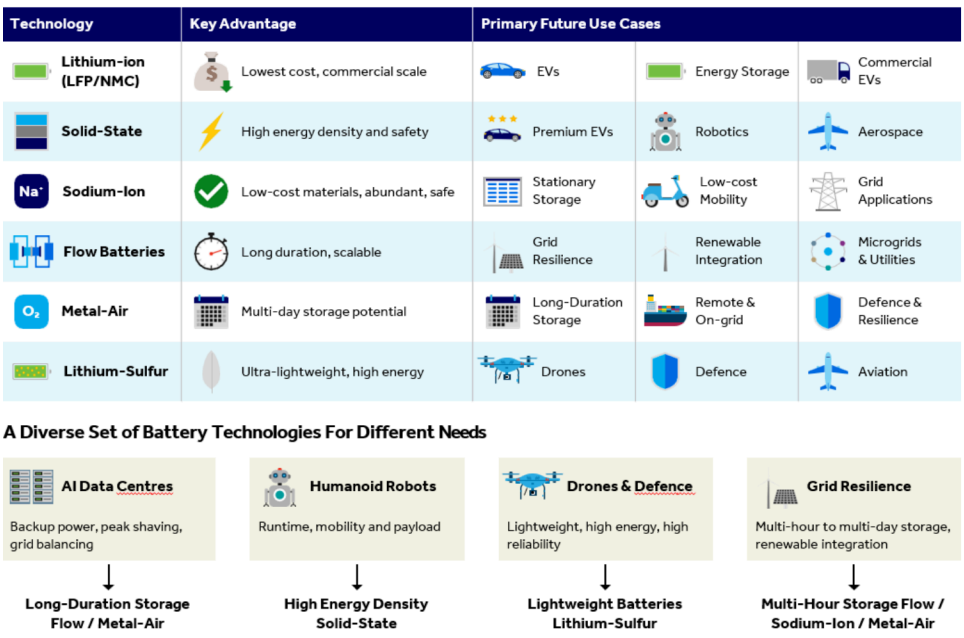
The growth of physical AI may also accelerate innovation cycles within the battery industry. Many first-generation robots and drones currently rely on battery technologies developed for larger end-markets such as electric vehicles and consumer electronics. However, as

deployments scale and utilisation becomes increasingly important, operators may be willing to pay a premium for performance improvements that enhance productivity or mission capability. This could support faster adoption of next-generation battery technologies, including silicon-based anodes and other advanced materials that offer advantages in charging speed, weight and energy density. Unlike EVs, where battery selection is often driven by cost per kilowatt-hour, battery procurement for physical AI applications may increasingly prioritise operational performance and total value creation, potentially supporting the n of higher-specification battery technologies.

Expanding end-market demand supports new technologies

Battery technology has improved immensely in the past few decades, driven by the need to improve energy density, storage efficiency, safety features, fast charging capabilities and thermal management while simultaneously lowering costs. Several next-generation technologies have been under development for many years, with recent progress pointing to early stages of n. Evolving end-demand is accelerating the need for new technologies. The rise of physical AI and humanoid robotics introduces higher energy density requirements, while increasing focus on grid resilience and stability requires long-duration energy storage (LDES) solutions (Urgency to scale Long Duration Energy Storage (LDES)). These broadening performance requirements are creating a more complex demand profile, amplifying the need for new technologies alongside today's lithium-ion batteries (for more details see our primer Science of Tomorrow: Where next for EV battery technology?).

FIGURE 16. **Emerging battery technologies and their potential use cases**



Source: Barclays Research

Lithium-ion innovation: Pushing incremental improvements further

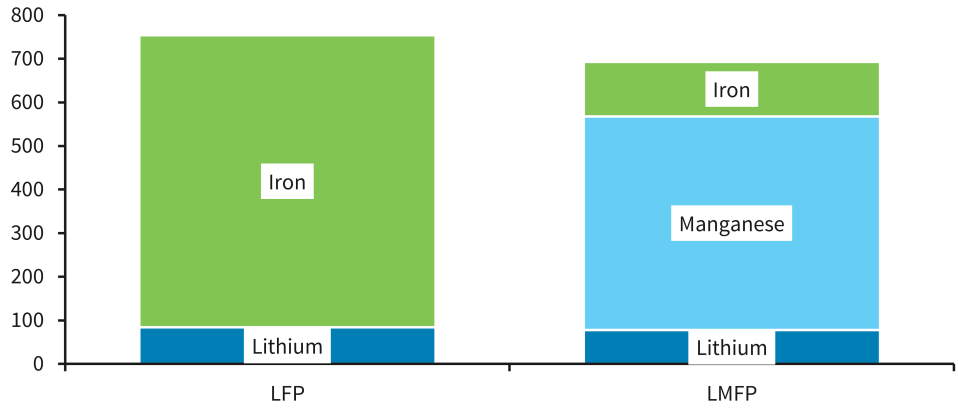
Cathode tweaks continue to improve performance

In the past decade, the most interesting aspects of battery innovation were largely cathode-driven, with the development of various nickel-containing (e.g. NMC 811) and nickel-free (e.g. LFP) chemistries. Companies continue to increase nickel content in batteries to improve energy density, especially with emerging demand from robotics and advanced air mobility

applications. For example, LG Energy Solution has stated that it is already supplying its high-nickel cylindrical batteries to over six major leading robotics players and is actively exploring applications in urban air mobility and aerospace.

Another ongoing technological trend is increasing usage of manganese in cathodes, most notably in LMFP (lithium manganese iron phosphate) batteries, which builds on LFP by partially substituting iron with manganese (Figure 17). Similar chemical approaches are also evident in LMR (lithium manganese-rich) and high-manganese NCM variants. The main benefit of higher manganese content is the ability to improve energy density at a relatively low incremental cost. As such, we view these chemistries as a bridge between LFP and NCM, offering a balance of cost, safety and performance. Since our deep-dive, such batteries have already been deployed (e.g. CATL's LMFP batteries used in Chery/Huawei car models).

FIGURE 17. Estimates for demand for 1MWh cathode material (kg)



LFP metal proportions taken from Bloomberg. LMFP metal proportions are Barclays estimates based on patent analysis.
Source: Bloomberg, Barclays Research

System-level improvements across anode, electrolyte and engineering also critical

Fast-charging has become a key battleground in EVs, particularly in LFP batteries. Fast-charging comes from system-level improvements across the electrolyte (higher conductivity), electrode materials (cathode coatings and anode surface engineering) and pack design and engineering (including thermal management). Significant progress has been made in fast-charging in recent years by companies such as CATL and BYD. CATL's Shenxing 3rd generation announced in 2026 claimed a 10-98% charge in less than 7 minutes. Similarly BYD's Blade 2.0 battery demonstrated 10-70% charge in ~5 minutes.

Progress has been steady in silicon anodes too, with increasing commercial traction. Silicon offers significantly higher theoretically energy capacity but technological challenges remain (e.g. volumetric expansion and stability). Recent developments, such as Sila opening its silicon anode facility last year, highlight the commercial progress being made. The $1.4bn loan commitment received by Sila from the Pentagon is also a notable milestone, given the US Department of Defense cited defence applications as one of the reasons for the investment. Given remaining technological hurdles, though, we expect adoption of silicon anodes to remain gradual, with silicon increasingly used as a blend within existing graphite anodes rather than an immediate replacement.

Beyond conventional lithium-ion: Solving the performance gaps

As performance requirements broaden due to evolving end-market applications, conventional lithium-ion batteries will struggle to meet the full spectrum of needs. Delivering a step-change in energy density and safety will require the adoption of new technologies, with five key next-gen technologies explored below (summarised in Figure 18). That said, given the significant

advances and scale that have been achieved in lithium-ion, we view these technologies as complementary to conventional lithium-ion, addressing use-cases where lithium-ion is constrained, rather than as structural replacements.

FIGURE 18. Five key emerging technologies

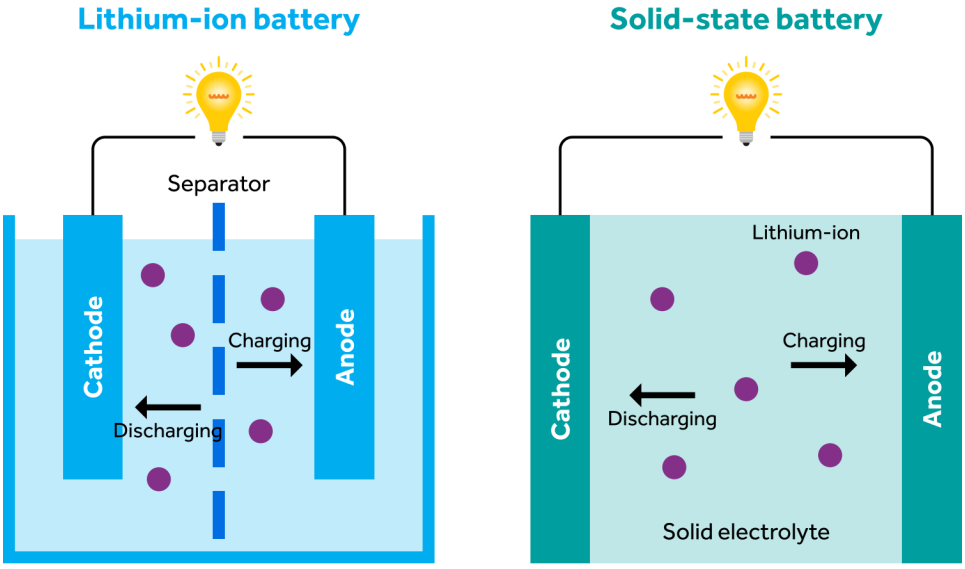
	Solid-state	Sodium ion	Redox flow	Metal-air	Lithium-sulfur
Main applications	• Premium EVs • High-end uses e.g. robotics, advanced air mobility	• Cost-sensitive mobility (EVs, 2-3 wheelers) • Stationary energy storage	• Long-duration energy storage	• Ulta long-duration energy storage (iron-air) • Medical devices (zinc-air)	• Aerospace and defence
Advantages	• Very high energy density • High safety	• Abundant materials (sodium) • High safety • Potentially lower cost	• Long lifespan • High safety • Scalability	• Multi-day duration • Low cost materials	• Very high energy density • Abundant materials (sulfur)
Disadvantages	• Scalability still being proven	• Lower energy density	• Low energy density • High upfront costs	• Short cycle life • Lower efficiency	• Short life cycle • Stability challenges
Technology readiness	Pilot-stage	Early commercial	Commercial (mostly vanadium flow)	Pilot-stage (mostly iron-air)	Pilot-stage
Examples of companies involved	Ganfeng Lithium*, Idemitsu Kosan*, QuantumScape*, ProLogium, Solid Power*	CATL**, BYD**, HiNa, Tiamat, Hithium Energy Storage, Altris	Invinity*, Sumitomo Electric*, Tharisa*, VRB Energy, ESS Tech*	Form Energy, Abound Energy* (both for iron-air)	Solidion Technology*, Lyten, LG Energy Solution*

Companies listed are private unless otherwise indicated: *indicates public but not covered by Barclays analysts. **indicates public and covered by Barclays analysts. The rest of the companies are private companies.
Source: Barclays Research

1) Solid-state: Advancing steadily, with humanoid robotics emerging as an early adopter

Often called the 'holy grail' of batteries, the biggest difference between solid-state batteries and conventional lithium-ion batteries is the type of electrolyte material used (Figure 19). While the majority of lithium-ion batteries use a lithium salt in organic solution, solid-state batteries use (as the name suggests) solid electrolytes. Solid electrolyte materials can broadly be classified as polymer, oxide, or sulfide, each boasting their own advantages and disadvantages. Crucially, one reason why solid-state batteries can boast such high energy density is the enabling of different anode materials to be used, such as lithium anodes or 'anode-less' batteries. For more detail on solid-state battery technology see our primer Science of Tomorrow: Where next for EV battery technology?.

FIGURE 19. **Conventional lithium-ion battery vs. a solid-state battery**



Source: Barclays Research

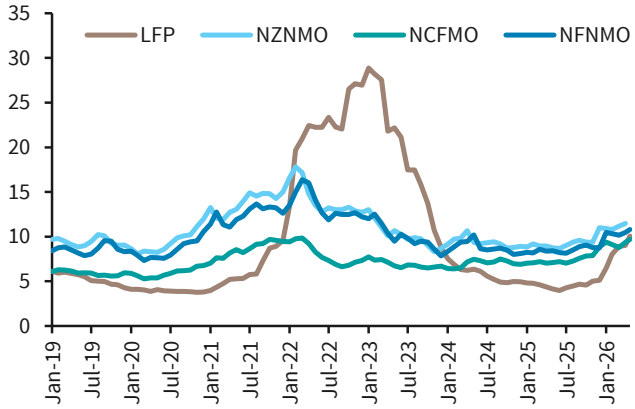
The progress made in n of solid-state batteries is encouraging, with clear advancements in recent years. For EVs, we see limited medium-term disruption risk to incumbent technologies given elevated manufacturing costs and complexity. Early adoption in this market is likely to be concentrated in premium applications.

In contrast, we see more a more compelling demand driver emerging from physical AI and humanoid robotics. These use-cases require high energy density due to space constraints and a need for smaller batteries. Furthermore the industry is less cost-sensitive at this stage of market development, enabling earlier adoption of cutting-edge batteries. Xpeng's 'IRON' humanoid robot, unveiled last year and scheduled for commercial sales in 2027, is already powered with an all-solid-state battery. Meanwhile battery producer LG Energy Solution has discussed developing anode-less solid-state battery technology with an initial focus on humanoid robots, indicating potential for initial deployment around 2030.

2) Sodium-ion: Raw material and safety benefits boosting adoption

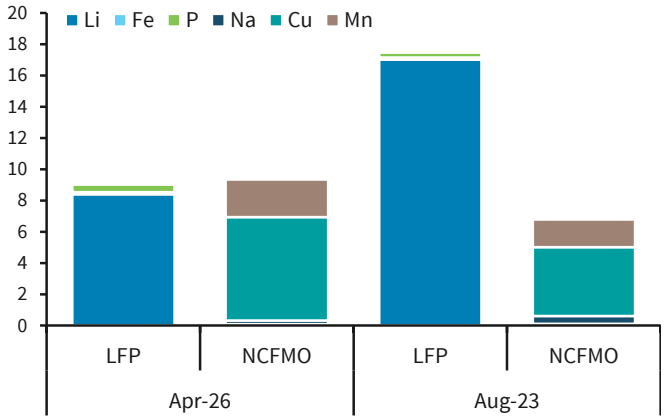
The main advantages of sodium-ion lie in its raw materials usage and safety benefits. Proponents of the technology highlight improved lower costs due to the abundance of sodium relative to lithium. However, our analysis of cathode costs has indicated that, while cost benefits can exist, they are not consistent (Figure 20 and Figure 21), and are even less evident when looking at the broader battery-pack level. The anode supply chain is also more nascent and remains heavily concentrated in China, as sodium-ion batteries are better suited to hard carbon anodes instead of graphite. For more on sodium-ion technology see Science of Tomorrow: Where next for EV battery technology?, and for more on the supply chain see Climate Tech: From decarbonisation to critical infrastructure.

FIGURE 20. Lithium-ion (LFP) cathode material costs vs. different sodium-ion (NZNMO, NCFMO, NFNMO) cathode material costs ($/kWh)



Source: W. Zuo et al., Acc. Chem. Res. 2023, 56, 284-296, Bloomberg, Barclays Research

FIGURE 21. Cathode material costs over time split by metal ($/kWh)



Source: Bloomberg, Barclays Research

Tecchnological progress since our deep-dive has been encouraging, though near-term applications remain limited to stationary energy storage and low-cost mobility. In energy storage, we see competitive pressures from second-life EV LFP batteries being repurposed in stationary storage which benefit from lower upfront costs vs. new sodium-ion batteries. That said, the safety angle should not be overlooked, particularly given fire risks with lithium-ion batteries (e.g. the high-profile Moss Landing BESS fire in California, 2025). Furthermore, while scepticism on sodium-ion's energy density vs. LFP may be warranted for now, the gap is narrowing. We note that early scepticism of LFP chemistries followed a similar trajectory, with subsequent engineering improvements paving the way for mass-market adoption. Assuming performance improves, we see sodium-ion as a risk to LFP if it can deliver a clear cost advantage and, if not, where improved safety is critical.

3) Flow batteries: A strong fit for LDES but high upfront costs

Flow batteries differ fundamentally from conventional lithium-ion batteries in how the electrolyte is stored. While a conventional lithium-ion battery has a fixed cell structure – an anode and cathode separated by an electrolyte in between – a flow battery stores the electrolytes in external tanks, separating the energy storage (the electrolyte tank) from the power output (the cell). This allows easy scaling by increasing electrolyte volume, making flow batteries particularly suited to long-duration energy storage (LDES). The most established flow battery being developed is vanadium flow (which uses vanadium in sulfuric acid as the electrolyte). Other alternatives include iron flow and organic flow, which use different electrolyte chemistries.

The difference in battery structure from lithium-ion batteries also causes a significant divergence in materials intensity. In flow batteries, energy is directly proportional to the volume of electrolyte. Because energy density is relatively low, very large volumes of electrolyte are needed per unit of energy. For example, in vanadium redox flow batteries the electrolyte makes up a major portion of overall system cost (30-50%). But, unlike lithium-ion batteries, there is no degradation of the electrolyte during use and so high upfront material demand is partly offset by minimal replacement needs over time.

Flow batteries are one of the more credible alternatives to lithium-ion for long-duration energy storage with the market already starting to scale given rising demand from data centres and broader grid-resiliency, but adoption will ultimately depend on cost competitiveness, scalability

and operational reliability. Early projects will be key to validating the commercial case and we don't think there is a one-size-fits-all LDES solution (Urgency to scale Long Duration Energy Storage (LDES)).

4) Metal-air batteries: Iron-air deployments will demonstrate readiness in LDES

Metal-air batteries differ fundamentally from conventional lithium-ion batteries in that the cathode material is not stored within the cell. While lithium-ion batteries contain both electrodes internally, metal-air systems use a metal anode (e.g. zinc or iron) and draw oxygen from ambient air as the cathode reactant, enabling much higher theoretical energy density by reducing the amount of stored material within the system.

There are several metal-air chemistries under development. Zinc-air is the most mature, benefiting from low-cost and abundant materials. These batteries are already widely used in hearing aids and other medical devices such as continuous glucose monitors. Iron-air systems are emerging for ultra-long-duration applications, targeting multi-day storage at low cost. However, performance is constrained by challenges around air electrode design, electrolyte stability and metal degradation, which impact efficiency and cycle life.

2026 has shaped to be an important year for iron-air batteries, with the first wave of initial deployment beginning to demonstrate the technology's potential (see Urgency to scale Long Duration Energy Storage (LDES) section for examples). While significant debate remains over the role of batteries in providing multi-day energy storage, particularly given the infrequent nature of such events and uncertain revenue opportunities, these early projects should provide valuable insight into the technology's real-world performance. We therefore see successful execution and operation of these initial deployments as a key milestone to watch.

5) Lithium-sulfur: Compelling in defence applications but still early-stage

Lithium-sulfur (Li-S) batteries differ from conventional lithium-ion systems in their use of sulfur as the cathode material with a lithium metal anode. Sulfur is lightweight, abundant and low-cost, enabling significantly higher theoretical energy density than lithium-ion, which makes Li-S particularly attractive for applications where weight and energy density are critical. However, the chemistry introduces technological challenges that lead to capacity fade, as well as issues around lithium metal stability, cycle life and safety. As a result, most development has focused on niche applications (e.g. aerospace and defence).

From a materials perspective, Li-S batteries benefit from lower reliance on critical minerals such as nickel and cobalt, but this is offset by the need for more complex system design to manage degradation mechanisms. While the potential for high energy density is clear, achieving consistent performance over many cycles remains the key barrier to commercial scale deployment.

Lithium-sulfur offers a compelling long-term pathway for high energy density storage using low-cost and abundant materials, particularly as growing momentum in aerospace and defence applications drives demand for lightweight batteries in drones and advanced military systems. For example, Lyten (a lithium-sulfur battery company) announced an initiative in 2025 to support growing demand for high-endurance drones in defence, taking orders for its latest lithium-sulfur battery for drones. From a thematic perspective, our 2035 Thematic Roadmap highlighted Defence Tech as a 2026/7 trend in focus in light of heightened geopolitical tensions. However, the technology remains nascent relative to other alternatives.

Identifying opportunities in next-gen technologies

While conventional lithium-ion batteries continue to improve through advances in cathodes, anodes, electrolytes and system engineering, evolving end-use requirements are accelerating

the need for complementary technologies. Higher energy-density applications such as humanoid robotics, alongside growing demand for long-duration energy storage and grid resilience, are creating opportunities for a range of next-generation battery chemistries. Figure 2 and Figure 18 highlight a selection of public and private companies developing or commercialising key emerging battery technologies.

Battery recycling vital for supply security, but scaling challenges persist

An attractive proposition for mineral independence

There is a clear strategic rationale for lithium-ion battery recycling, both from a circularity perspective but also for critical mineral independence. For companies, vertical integration including recycling can allow much more control and oversight of supply chains, while for governments recycling reduces dependency on imports of critical minerals and allows countries to overcome geological constraints (Science of Tomorrow: EV Battery Recycling).

While consumer electronic battery recycling is already relatively established, helped in part by more accessible collection infrastructure, EV battery recycling is critical to provide larger volumes of minerals such as lithium, cobalt and nickel, and therefore should be the primary focus for investors in the medium term, in our view (Electronic Waste: The Race for Critical Materials).

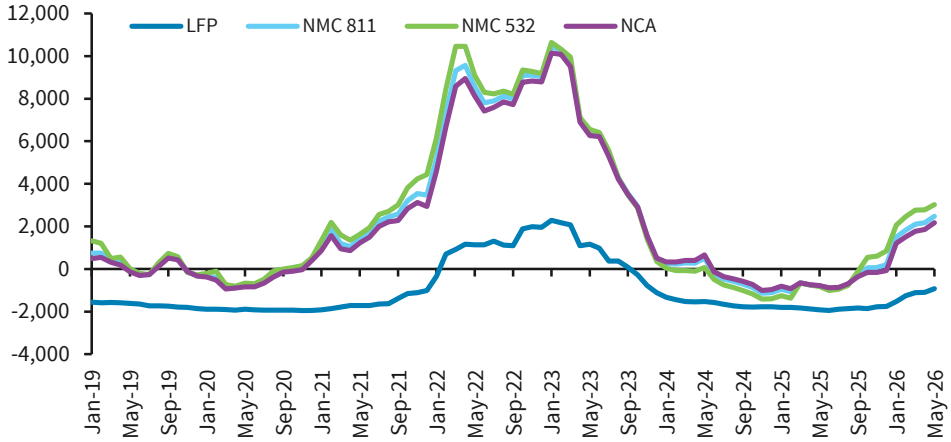
Scaling challenges need to be overcome

Despite the strategic rationale for large-scale EV battery, the industry has seen several hurdles in recent years that have led to project delays and cancellations. We think there have been two key problems contributing to this:

1) Feedstock availability: A combination of improving battery lifetime and a slowdown in EV adoption growth rates has led to a feedstock availability being pushed out by several years. For example, battery recycler Hydrovolt stated that longer battery lifetimes were delaying the company's capacity ramp-up, such that it processed only about a third of its 12,000 tpa target last year.

2) Recycling profitability: Low cost is one of the main factors that has driven LFP batteries to gain market share relative to other battery chemistries (Battery Recycling: Profitability and export risks in CRM supply chains), yet this also means the value proposition of recycling LFP batteries is much less attractive given the lower value of the cathode metals, as indicated in Figure 22. In response to this and growing demand for energy-storage batteries, recyclers such as Redwood Materials have pivoted their strategies to focus on converting end-of-life EV LFP batteries to stationary energy storage applications, a trend we expect to continue.

FIGURE 22. **Estimated recycling profit in Europe, $/tonne**



Per tonne of input scrap
Source: BloombergNEF, Bloomberg, Barclays Research

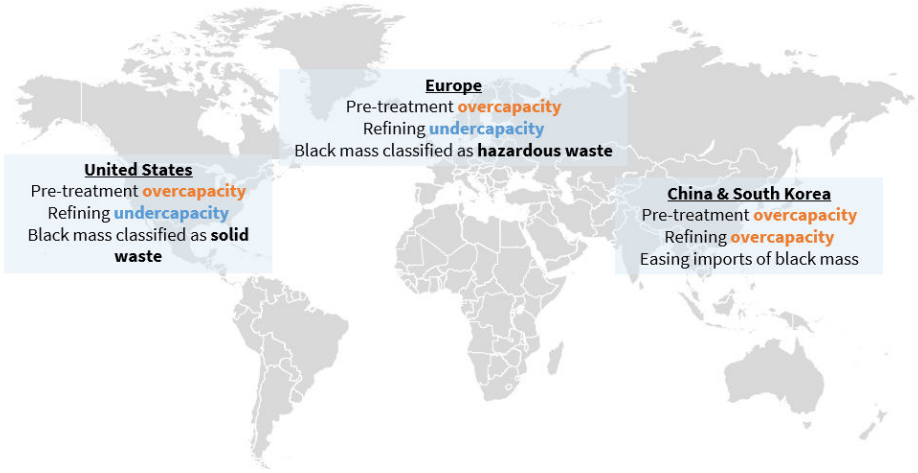
Beyond EVs, recycling end-of-life ESS batteries presents similar (if not more severe) difficulties as the chemistry mix is even more exposed to low-value LFP and there is greater variation in format and degradation of the batteries themselves, which will further require more complex sorting.

Regulatory catalysts provide reason to be hopeful

Despite some of these challenges, we see potential catalysts on the horizon that should support the industry, especially in the EU. The EU battery passport is due to be introduced in 2027 allowing greater transparency over supply chains and sourcing. In the medium term, the EU Batteries Regulation, which came into force in 2023, will introduce minimum recycled content in batteries from 2031. In the US, the EPA is developing a national Extended Producer Responsibility (EPR) framework to legislate the management of end-of-life EV batteries. In the meantime, states such as California and Washington have developed their own EPR frameworks around batteries, which also should facilitate feedstock collection infrastructure when larger volumes of batteries reach end-of-life.

Finally, we have noted before that oversupply of processing capacity but undersupply of hydrometallurgy capacity in the EU and US has led to the risk of black mass being exported to China where the battery recycling industry is more established (Figure 23, discussed in more detail in Battery Recycling: Profitability and export risks in CRM supply chains). Moves from the EU to classify black mass as hazardous waste should limit these exports and therefore help retain feedstock for new hydrometallurgy plants that convert black mass to battery-grade salts. Looking ahead, regulatory demand signals and clearer volumes visibility will be key for the industry to reach an inflection point, in our view.

FIGURE 23. **Regional battery recycling capacity dynamics**

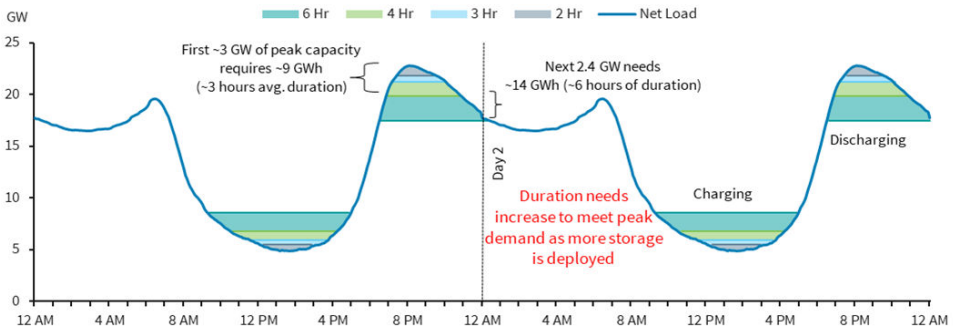


Source: Rho motion, Barclays Research

Urgency to scale Long Duration Energy Storage (LDES)

Energy storage duration needs are increasing. Energy storage will play a critical role in enabling a clean, reliable and affordable grid (Utility-scale Energy Storage: For Green Grid, Batteries Required; Energy Storage Summarized in 16 Slides). However, as the share of intermittent renewable generation (solar, wind) rises and storage capacity scales, the shape of the net load curve is evolving – the peak height compresses, but its duration extends. As a result, incremental storage capacity must serve a broader window of demand, increasing duration requirements. As illustrated in Figure 24, this dynamic reinforces the need for longer-duration storage to maintain reliability during extended periods of peak demand and, ultimately, to help firm variable renewables toward 24/7 carbon-free supply. According to the US Department of Energy, long duration storage (≥10 hours) needs could reach 225-460 GW by 2050 in order to provide clean, firm capacity alongside high shares of variable renewables.[13]

FIGURE 24. **Illustration of marginal storage duration requirements to meet peak power demand based on California ISO April 2021 net load curve**



Source: California ISO, Barclays Research

Lithium-ion will only go so far. Lithium-ion (Li-ion) has benefited from the scale of the EV market, driving improvements in cost, reliability, energy density and round-trip efficiency, making it the most commercially mature and 'bankable' storage technology today. However, Li-ion faces structural limitations at longer durations. Power and energy costs are tightly coupled – extending storage duration (e.g. from 4 to 8 hours) requires a near-linear increase in cell capacity, resulting in a high marginal cost of additional energy ($/kWh). Put differently, Li-ion

[13] US Department of Energy: Pathways to Commercial Liftoff: Long Duration Energy Storage

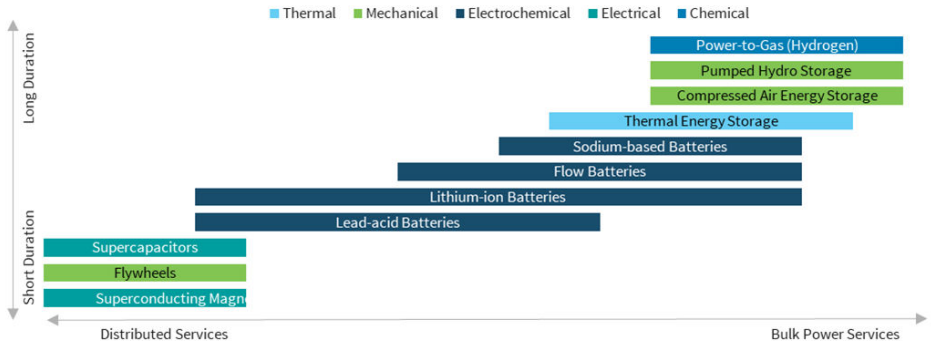
does not exhibit favourable economics of duration scaling. While its high energy density is critical for mobile applications, it is less relevant in stationary storage, where footprint constraints are typically less binding.

By contrast, many alternative storage technologies decouple power ($/kW) and energy ($/kWh) costs, enabling significantly lower marginal cost of duration. Li-ion is therefore likely to remain dominant in short-duration applications (e.g. ≤4 hours), supported by continued cost declines and strong performance characteristics. However, its relatively high marginal energy cost increasingly limits competitiveness in longer-duration (e.g. 6-8+ hour) use cases. In parallel, interest in alternative chemistries and architectures is rising, driven by considerations around safety, degradation, cycle life, supply chain concentration and sustainability.

This creates urgency for long-duration energy storage (LDES). Technologies that can deliver low-cost duration are increasingly critical to avoid constraining renewable penetration and overall grid decarbonisation. Historically, pumped hydro storage (PHS) has been the primary LDES solution. PHS operates as a large-scale 'water battery', storing energy by pumping water to an elevated reservoir and releasing it through turbines when needed. It offers very low marginal duration cost, as extending storage typically requires increasing reservoir capacity rather than duplicating generation equipment. However, new PHS development has been limited in most markets due to high upfront capital costs, long development timeline, and siting and environmental constraints.

No one-size-fits-all LDES solution. A diverse set of LDES technologies is emerging across multiple stages of n. Broadly, storage technologies fall into four primary categories: mechanical (e.g. pumped hydro, compressed air), electrochemical (e.g. batteries), chemical (e.g. hydrogen and derivatives) and thermal (e.g. heat-based storage). Each technology exhibits distinct trade-offs across duration, cost structure, efficiency, response time, operational flexibility and siting requirements. As a result, the LDES landscape is likely to remain heterogeneous, with different technologies serving specific grid needs rather than a single dominant solution.

FIGURE 25. **Select energy storage technologies by duration & application**



Source: National Renewable Energy Lab, Barclays Research

Examples of recent LDES advancements include:

- **Form Energy's** (private) iron-air battery system – designed to deliver ~100 hours of storage – has been selected by utilities including Xcel Energy and Great River Energy to provide reliability during prolonged renewable shortfalls and support clean firm capacity. The company has secured hyperscaler-linked deployments, including a Google-backed 300 MW / 30 GWh (~100-hour) system supporting a data centre, and a separate agreement with Crusoe for 12 GWh of multi-day storage for AI data centres. Form is scaling manufacturing through its

first US facility in West Virginia, intended to enable multi-GW deployment over time, alongside >$800 million in cumulative fundraising, including a ~$405mn Series F in 2024.

- **Energy Dome** (private) is advancing n of its CO_2-based energy storage technology (10–24 hours), with early commercial agreements in Europe and growing traction in the US, positioning as a lower-cost alternative to lithium-ion for medium-duration applications. The company has expanded into hyperscaler-adjacent markets through a strategic commercial partnership and investment from Google, supporting carbon-free energy deployment at scale, and an MOU with NUAI to deploy storage at a >1 GW Texas AI data centre site, reflecting early adoption in data centre power solutions.
- **Hydrostor** (private) has secured major contract awards for its Advanced Compressed Air Energy Storage (A-CAES) technology, including multi-hundred-megawatt projects in California and Australia, targeting long-duration (8-24+ hour) firm capacity applications. The company is backed by significant equity investment and project-level financing to advance a multi-billion-dollar development pipeline.
- **Eos Energy Enterprises** (EOSE, not covered) is scaling US-based manufacturing of its zinc-based battery systems (3-12 hours), supported by DOE financing and additional capital raises (~$336mn equity and convert issuance in mid-2025). The company has moved into data centre power solutions through a joint development agreement with Turbine-X, targeting up to ~2 GWh of deployments for AI/hyperscale projects using behind-the-meter 'bring-your-own-power' configurations.
- **ESS Tech** (GWH, not covered) is deploying iron-flow batteries (4-12+ hours) across utility and commercial projects in the US and Europe, emphasising long cycle life and minimal degradation for daily cycling applications, while expanding manufacturing capacity in Oregon with government and strategic support.

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